Week 3 – Application selection for performance testing

CPU Test

To test the CPU, I used sysbench which creates an intense load on the CPU. It achieves this by running multiple heavy operations such as multi-threaded computation loops and integer mathematical operations. This pushes the CPU to use 100% making it ideal to be tested for things such as thermal throttling and load balancing. Sysbench is good for this because it is designed for benchmarking and is highly configurable.

```
katana@katana-VirtualBox:~$ sysbench cpu --cpu-max-prime=20000 --threads=6 --time=60 run
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 6
Initializing random number generator from current time


Prime numbers limit: 20000

Initializing worker threads...

Threads started!

CPU speed:
    events per second:   996.73

General statistics:
    total time:                          60.0079s
    total number of events:              59813

Latency (ms):
         min:                                    1.55
         avg:                                    6.02
         max:                                 3002.88
         95th percentile:                       12.08
         sum:                               359948.54

Threads fairness:
    events (avg/stddev):           9968.8333/549.66
    execution time (avg/stddev):   59.9914/0.01

katana@katana-VirtualBox:~$
```

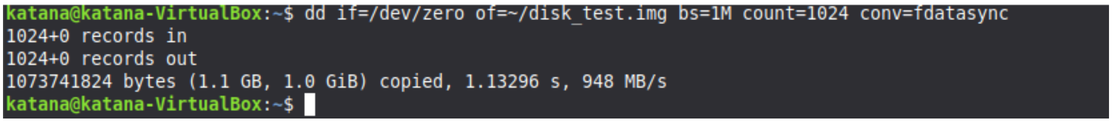
RAM

For RAM intensive testing, I used stress -ng which selects large blocks of memory and performs read/write operations to stress the RAM. This is necessary because it allows me to emulate scenarios where RAM is saturated, allowing you to analyse memory fragmentation and system stability.

```
katana@katana-VirtualBox:~$ stress --vm 1 --vm-bytes 1G --timeout 60
stress: info: [13366] dispatching hogs: 0 cpu, 0 io, 1 vm, 0 hdd
stress: info: [13366] successful run completed in 60s
katana@katana-VirtualBox:~$ stress --vm 1 --vm-bytes 1G --timeout 60
stress: info: [13373] dispatching hogs: 0 cpu, 0 io, 1 vm, 0 hdd
^C
katana@katana-VirtualBox:~$ stress --vm "$(nproc)" --vm-bytes 1G --time
stress: info: [13380] dispatching hogs: 0 cpu, 0 io, 2 vm, 0 hdd
stress: info: [13380] successful run completed in 60s
katana@katana-VirtualBox:~$
```

Disk

To test the disk, I used dd. This is a low level utility which is used to read and write raw data blocks directly to storage devices. Dd is useful here because it is simple and reliable as well as efficient when it comes to showing results



```
katana@katana-VirtualBox:~$ dd if=/dev/zero of=~/disk_test.img bs=1M count=1024 conv=fdatasync
1024+0 records in
1024+0 records out
1073741824 bytes (1.1 GB, 1.0 GiB) copied, 1.13296 s, 948 MB/s
katana@katana-VirtualBox:~$
```

Network

To test network, I used iperf3 which generates high-throughput TCP/UDP traffic bandwidth, jitter and packet loss